SNAPDRAGON® AI LAB BUILD & PRESENT CHALLENGE

Vaani Khata

Speak your shop's ledger. Hindi, English or Hinglish.
Transcribed on the Snapdragon NPU. Nothing leaves the laptop.

Individual entry · Built and measured on a Snapdragon X laptop (Galaxy Book4 Edge, Windows 11 ARM64)

The khata is still on paper

1

Paper is fragile

Kirana shopkeepers track credit (udhaar) in handwritten books. Entries get lost, disputed, or forgotten.

2

Apps assume typing and internet

Cloud ledger apps need connectivity, an account, and typing in English menus during a rush.

3

Customer data leaves the shop

Who owes what is sensitive. Cloud apps upload it. Voice makes that worse.

Talk to it like you talk to your helper

1

Speak

Tap once. "Ramesh ne 500 rupaye udhaar liye."

2

Whisper on NPU

AI Hub Whisper-Small, Hexagon NPU via QNN

3

Understand

Entry type, person, amount, item, spoken in Hindi or English

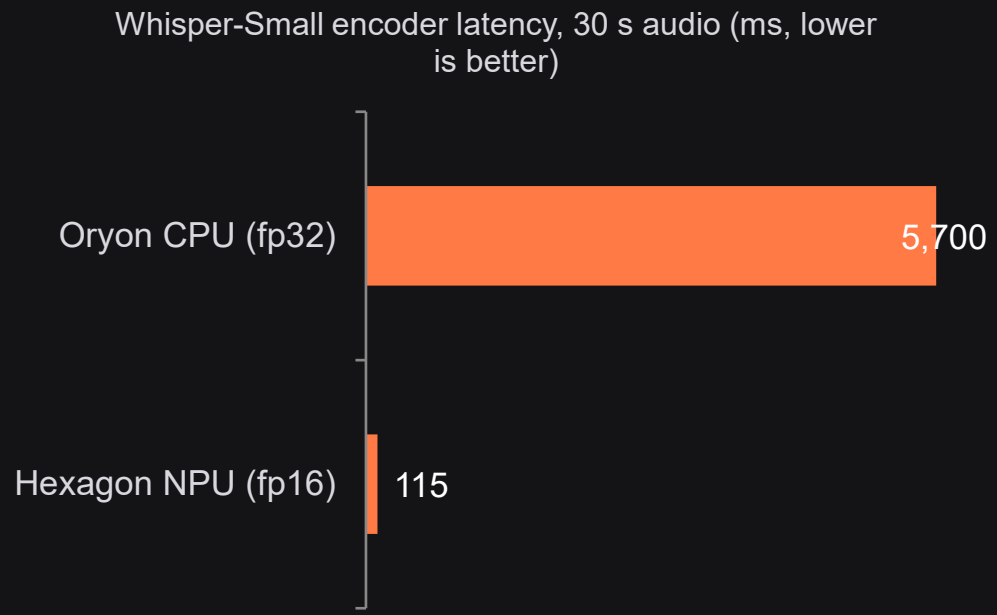
4

Khata

Local SQLite. Balances update; ask "Ramesh ka kitna baaki?"

Auto-detects Hindi vs English. Hindi and English spellings of one name (रमेश / Ramesh) merge into one customer.

Why the NPU: measured on this laptop



~50x

faster than the CPU (48.6x and 50.9x in two runs; NPU steady at ~115 ms)

16x

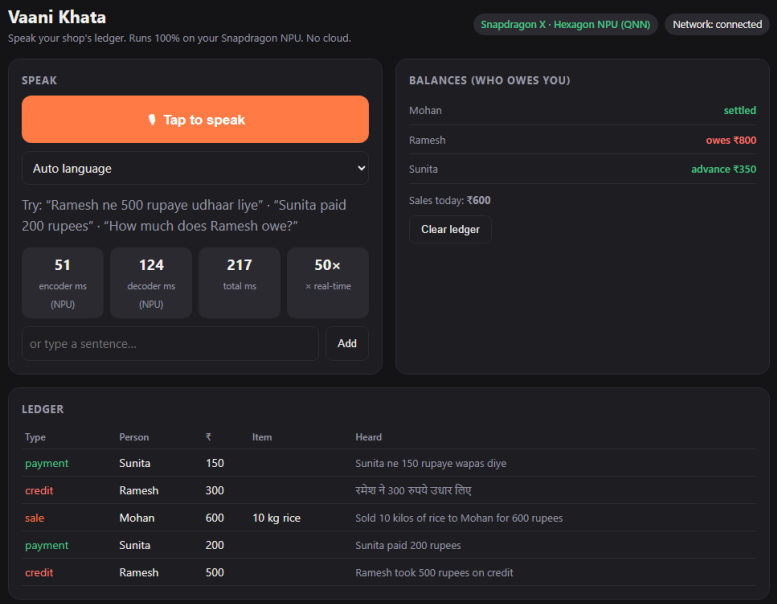
real-time: a 10.8 s spoken clip transcribed end to end in ~0.65 s

0 bytes

of audio or ledger data leave the device

Baseline: open whisper-small encoder (same architecture) on the ONNX Runtime CPU EP. Full runs and caveats in docs/benchmark.md.

The working app



Tap to speak

Mic in the browser, audio processed by a localhost Python server. No account.

Live ledger and balances

Credit, payment and sale entries with the exact words heard for each.

Ask by voice

"How much does Ramesh owe?" answers from the local database.

Offline badge

Works with Wi-Fi off. Latency shown on screen every time.

Built to actually be deployed

Runs where shops are

Any Snapdragon X Windows laptop. One setup script downloads the AI Hub model. No GPU, no cloud, no monthly bill.

Accessible by voice

Hindi, English and Hinglish today. No typing, no menus. Hindi and English spellings of a name merge into one customer.

Private by design

Server binds to 127.0.0.1. Data lives in one SQLite file on the shopkeeper's own machine.

Next: Llama 3.2 3B on the NPU for free-form sentences, GST-ready sales summaries, and more Indian languages.

What is real today

Working, measured on-device

- Whisper-Small encoder + decoder on the Hexagon NPU (QNN)
- Speech to ledger entries, balances, spoken balance queries
- Hindi and Hinglish speech: 6 of 6 test sentences correct (Whisper-Base: 2 of 6)
- Reproducible benchmark script and logged results

Known limits

- Sentence parsing is rule-based, so unusual phrasing can be missed
- Hindi tested only with synthetic voices; real shop noise and accents will be harder
- CPU baseline is the same architecture from an open ONNX export, not the identical file

Vaani Khata: private, instant voice bookkeeping, powered by the Snapdragon NPU.